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2027 CYCLE

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CUET UG

Mathematics

Inferential Statistics

Sampling, Parameters, Central Limit Theorem and the t-Test · Section B2 VI · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) 2026 syllabus — Mathematics/Applied Mathematics (319), the current 2027 baseline)	Skills the questions test
Population and sample	"Define Population and Sample. Differentiate between population and sample. Define a representative sample from a population. Differentiate between a representative and a non-representative sample." (Section B2 Applied Mathematics Unit VI)	Naming groups correctly; spotting biased sampling in a described study.
Sampling designs	"Draw a representative sample using simple random sampling. Draw a representative sample using a systematic random sampling." (Section B2 Applied Mathematics Unit VI)	Running an SRS draw; computing $k = \frac{N}{n}$ and listing a systematic sample.
Parameter, statistic and inference	"Define Parameter with reference to Population. Define Statistics with reference to Sample. Explain the relation between Parameter and Statistic. Explain the limitation of Statistic to generalize the estimation for population. Interpret the concept of Statistical Significance and Statistical Inferences." (Section B2 Applied Mathematics Unit VI)	Greek-vs-Latin bookkeeping; honest limits of generalising one sample.
Central Limit Theorem and the t-test	"Central Limit Theorem: Explain the relation between Population-Sampling Distribution-Sample. t-Test (one sample t-test for a small group sample): Define a hypothesis. Differentiate between Null and Alternate hypothesis. Define and calculate degree of freedom. Test Null hypothesis and make inferences using t-test statistic for one group." (Section B2 Applied Mathematics Unit VI)	Standard error arithmetic; df; a full five-step t-test verdict.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every

correct answer and −1 for every incorrect answer. The 2026 syllabus houses this unit as **Section B2 Applied Mathematics Unit VI** — the chapter where vocabulary carries the marks: most questions are definitions and verdicts, with one compact formula (the t-statistic) doing the numerical work.

Population and Sample — the Cast of Characters

The **population** is everyone (or everything) the study is about — all Class 12 students in a state, every bulb a factory makes. The **sample** is the subset you actually measure, because measuring whole populations (a **census**) is usually impossible or absurd. The bridge between them is the **representative sample**: a small group that mirrors the population's spread — same mix of urban and rural, same gender balance, same age texture — so that what happens in the sample plausibly happens outside it. A **non-representative** sample breaks that mirror: surveying only your coaching class about exam stress, or testing bulbs from a single machine, produces confident answers about the wrong group.

- **Bias is a sampling property, not a result property.** A convenient sample can look perfectly reasonable and still be structurally skewed — the flaw lives in **how** units were chosen.
- **Sampling frame:** the actual list you draw from. If the frame omits a subgroup, no cleverness downstream repairs it.

Two Sampling Designs — SRS and Systematic

Simple random sampling (SRS) gives every unit — and every combination — an equal chance: number the population, then draw by lottery or a random-number table. With replacement, a unit can return (SRSWR); without replacement, drawn units sit out (SRSWOR, the usual classroom default). **Systematic random sampling** trades lotteries for rhythm: compute the jump $k = \frac{N}{n}$ (population size over sample size), choose a **random start** between 1 and k, then take every k-th unit. $N = 100$, $n = 10$ gives $k = 10$; a random start of 7 yields 7, 17, 27, …, 97 — ten units spread evenly across the whole list, chosen by a single random act.

WORKED EXAMPLE — DESIGNING BOTH SAMPLES

A college has 400 students numbered 1–400; a study needs 40. Design an SRS and a systematic sample.

Solution. SRS: use a random-number table to draw 40 distinct numbers from 1–400 (without replacement), discarding repeats and out-of-range draws. **Systematic:** $k = \frac{400}{40} = 10$; choose a random start in 1–10 — say 6 — and take 6, 16, 26, …, 396. Both designs give every student a known, fair chance; the systematic version additionally guarantees the sample is spread across the entire list. The one discipline: the start must be **random** — always taking student 1 makes it a convenience sample wearing a formula.

Parameter vs Statistic — Greek and Latin

Bookkeeping with two alphabets: **parameters** describe the population and are written in Greek — μ the population mean, σ the population standard deviation. **Statistics** describe the sample and are written in Latin — $|x|$ the sample mean, s the sample standard deviation. The parameter is a fixed constant (the population has one true mean, whether or not we ever learn it); the statistic **wobbles** from sample to sample — draw again, get a different $|x|$. The inference deal: use the wobbling statistic as the **estimate** of the fixed parameter. The syllabus's named limitation is exactly the wobbling: one sample's $|x|$ is close to μ in the long run, but never guaranteed close **in this particular draw** — sampling error is the rent we pay for not measuring everyone.

THE BOOKKEEPING TABLE

$$\text{Population} \rightarrow \mu, \sigma \qquad \text{(parameters — fixed, usually unknown)}$$

$$\text{Sample} \rightarrow |x|, s \qquad \text{(statistics — computed, they vary)}$$

$$\text{Inference: } |x| \text{ estimates } \mu, \quad s \text{ estimates } \sigma$$

The Central Limit Theorem — Why Averages Behave

Draw many samples of size n and collect their means: that collection is the **sampling distribution of $|x|$** — the middle character between population and sample. The Central Limit Theorem gives its law: the means centre on μ (no bias), their spread is the **standard error** $\text{SE} = \frac{\sigma}{\sqrt{n}}$, and their **shape** turns normal as n grows — **even if the population itself is not normal**. Bigger samples squeeze the wobble: quadruple n and the standard error halves. That is the whole logic of "larger surveys are more reliable", written as one formula.

WORKED EXAMPLE — STANDARD ERROR IN ACTION

A population has $\mu = 60$ and $\sigma = 12$. A sample of $n = 36$ is drawn. Describe the sampling distribution of $|x|$, and find the probability that $|x|$ exceeds 62.

Solution. By the CLT: mean $\mu = 60$, standard error $\text{SE} = \frac{12}{\sqrt{36}} = 2$, and with $n = 36$ the shape is essentially normal. Then

$$z = \frac{62 - 60}{2} = 1, \quad P(|x| > 62) \approx P(Z > 1) \approx 0.1587.$$

About a 16% chance. Notice the population's own shape never entered the answer — the CLT lent the normality. Forgetting the $\sqrt{n}$ in the denominator is the classic slip: it would have doubled the z-score and wrecked the probability.

Hypotheses and the One-Sample t-Test

A **hypothesis** is a testable claim about the population. The **null hypothesis** H_0 is the status quo — "the mean is 50", "the machine is fine" — and it always carries the equality. The **alternate**

hypothesis H_1 is the doubt being proposed: $\mu \neq 50$ (two-tailed), or a strict $\mu > 50$, $\mu < 50$ (one-tailed). The test never "proves" anything true; it **rejects** or **fails to reject** H_0 at a pre-set **significance level** (usually 5%) — the accepted risk of a false alarm.

For a **small** sample with unknown σ, the **one-sample t-test** replaces the z-score:

$$t = \frac{|x| - \mu_0}{\frac{s}{\sqrt{n}}}, \quad \text{degrees of freedom} = n - 1$$

— the gap between the sample mean and the claimed mean, scaled by the standard error estimated with s. One degree of freedom is spent estimating the mean, hence $n - 1$. The verdict: if $|t|$ exceeds the critical table value for that df at 5%, the result is **statistically significant** — reject H_0; otherwise keep it.

WORKED EXAMPLE — THE FULL FIVE-STEP T-TEST

A vendor claims his packets weigh 50 g on average. A sample of 16 packets has mean 48 g and standard deviation 4 g. Test the claim at the 5% level (two-tailed).

Solution. Step 1: $H_0 : \mu = 50$; $H_1 : \mu \neq 50$. **Step 2:** standard error $\frac{s}{\sqrt{n}} = \frac{4}{4} = 1$, so $t = \frac{48-50}{1} = -2$. **Step 3:** df $= 16 - 1 = 15$. **Step 4:** the two-tailed 5% critical value for 15 df is 2.131. **Step 5:** $|-2| = 2 < 2.131$ — the result is **not** significant: fail to reject H_0. The sample is 2 g light, but with only 16 packets and this much wobble, that gap is within normal sampling noise. The evidence is too weak to indict the claim — and note the verdict language: never "H_0 is true", only "not rejected".

EVERYDAY EXAMPLE — TASTING THE SOUP

A cook stirs the pot (the population) and tastes one spoon (the sample). The spoon is a statistic: it wobbles — a bad scoop misses the chilli. The pot's true flavour is the parameter: fixed, but never tasted directly. A **representative** spoon means the soup was stirred before tasting; tasting from the surface only is a convenience sample — the salt verdict will be wrong no matter how carefully you sip. The t-test is the cook's discipline: "is this spoon different enough from what the recipe promises that I should blame the pot, or could any honest spoon wobble this much?" Small spoon, more wobble — that is why small samples need the gentler t-table instead of the confident z.

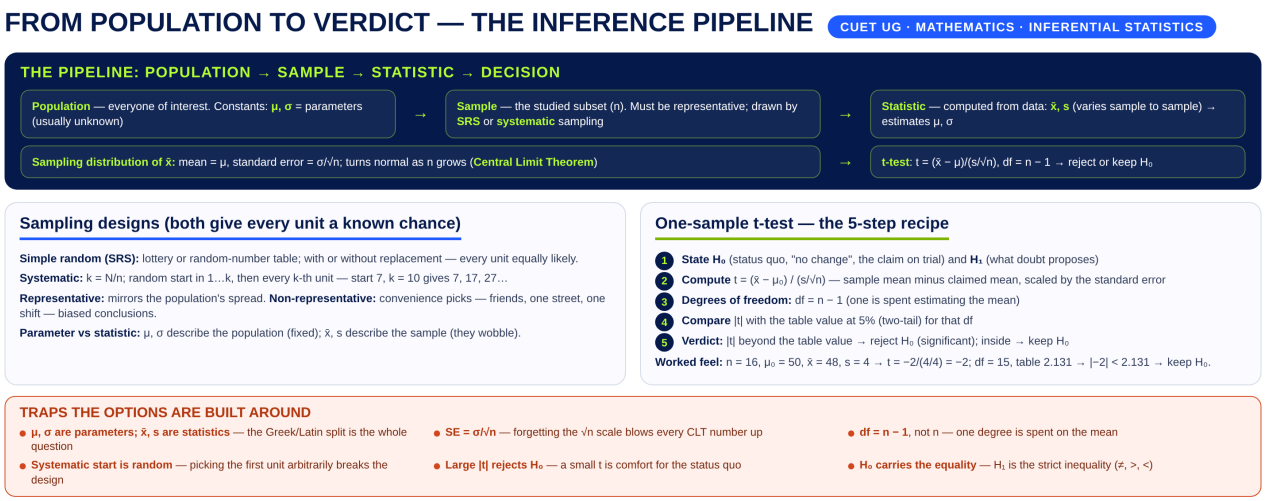


Figure 1: The inference pipeline: population to sample to statistic to decision, with the sampling designs, the CLT and the five-step t-test in place.

Formula Cheatsheet — the Whole Chapter on One Page

FORMULA CHEATSHEET — INFERENTIAL STATISTICS

Formula / rule	Letters mean	Fire it when
Population → parameters μ, σ	fixed, usually unknown	definition MCQs
Sample → statistics $\lvert x\rvert, s$	computed, they vary	definition MCQs
Systematic: $k = N/n$, random start, every k-th	population over sample size	design questions
SRS: random numbers / lottery, equal chance	WR vs WOR	design questions
Representative = mirrors the population's spread	bias lives in the choosing	critique stems
CLT: mean of $\lvert x\rvert$ is μ	no bias	sampling distribution
Standard error: SE $= \frac{\sigma}{\sqrt{n}}$	the $\sqrt{n}$ scale	every CLT number
Shape of $\lvert x\rvert$'s distribution → normal as n grows	population shape irrelevant	CLT logic

Formula / rule	Letters mean	Fire it when
Hypotheses: H_0 carries equality; H_1 is $\neq$, $>$, $<$	status quo vs doubt	setup stems
$\text{t} = \frac{\lvert x\rvert - \mu_0}{\frac{s}{\sqrt{n}}}$	gap scaled by SE	one-sample tests
$\text{df} = n - 1$	one spent on the mean	table lookup
Verdict: $\lvert t\rvert$ beyond table value $\rightarrow$ reject H_0	significant at 5%	final inference

Common Errors and Special Cases

COMMON ERRORS AND SPECIAL CASES — INFERENCE

The mistake	What is right	The trap it sets
Calling $\lvert x\rvert$ a parameter	Sample numbers are statistics; Greek letters are parameters	Greek/Latin swap options
Systematic start chosen as 1	The start must be random in $1..k$	Convenience sampling in disguise
SE computed as $\frac{\sigma}{n}$	$\frac{\sigma}{\sqrt{n}}$ — square root scale	Doubled z-scores, wrong probabilities
Normal shape assumed for any n	The CLT grows it as n increases	Small-sample overconfidence
$\text{df} = n$	$n - 1$ — one degree spent on the mean	Wrong critical value, flipped verdict
"H_0 is proved true"	Verdicts: reject or fail to reject	Proof-language options
Rejecting on a small $\lvert t\rvert$	Small t = comfortable status quo	Reversed verdict stems
Reading a one-tailed table for two tails	Match the tail to H_1's shape	Half the risk, wrong table

Sixty-Second Self-Test

Statement	Verdict
σ is a statistic.	False — Greek letter, population parameter
Sampling every 10th student after a random start is systematic sampling.	True — the design's signature
The standard error shrinks as n grows.	True — $\frac{\sigma}{\sqrt{n}}$
The CLT needs the population to be normal.	False — it builds the normality itself
df for n = 20 is 20.	False — 19, one spent on the mean
A huge $\lvert t \rvert$ comfortably keeps H_0.	False — it rejects it

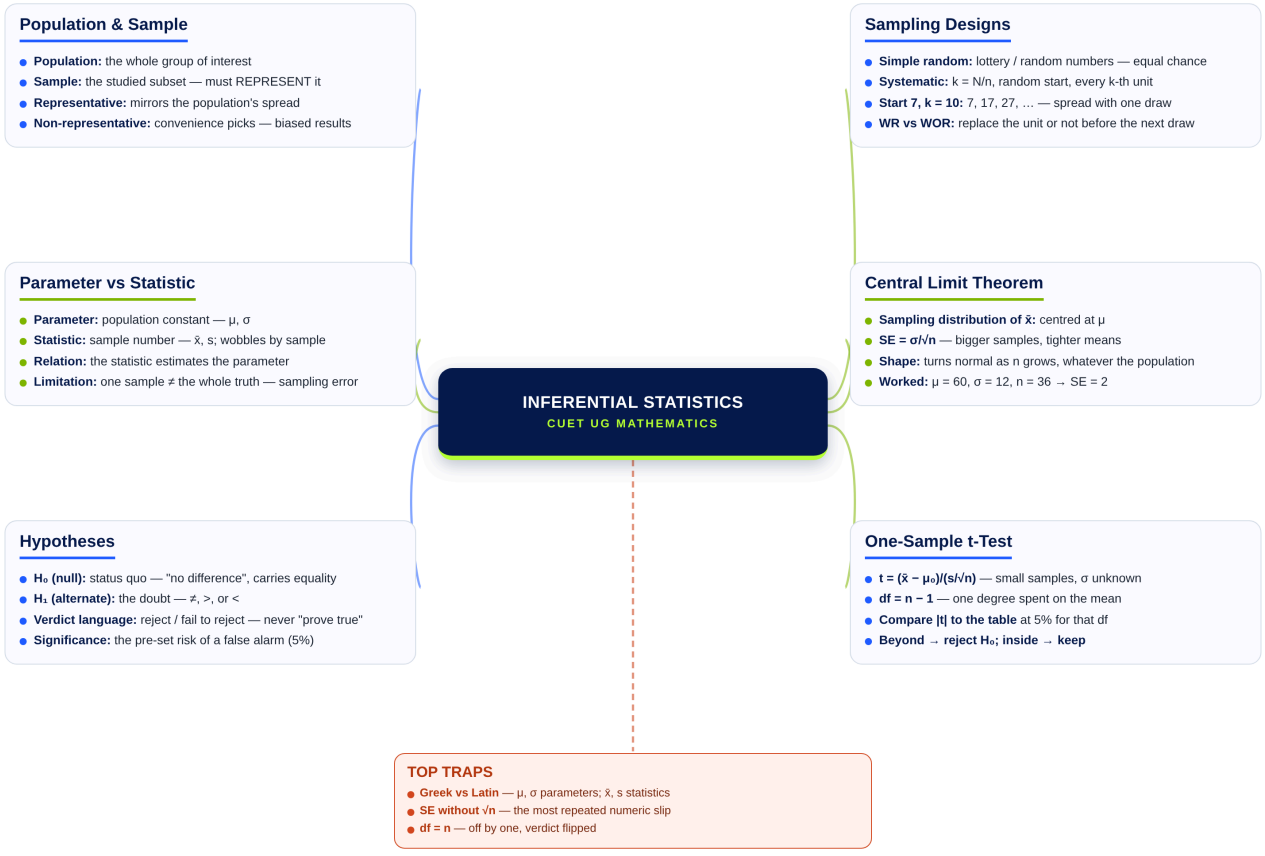


Figure 2: The chapter in one look: population and sample, designs, parameters, CLT, hypotheses and the t-test — traps included.

Practice Questions — CUET UG Pattern (10 MCQs)

Ten questions in the CUET UG style. Each correct answer earns +5; each incorrect one costs −1. Vocabulary first, formula second — that is the natural order of this unit.

Q1. The average height computed from all 2,000 students of a school is:

(A) a statistic
(B) a parameter
(C) a standard error
(D) a sampling frame

Q2. In systematic sampling with $N = 200$ and $n = 20$, the sampling interval k is:

(A) 20
(B) 200
(C) 40
(D) 10

Q3. A researcher surveys only the students present in one classroom because it is convenient. The sample is:

(A) simple random
(B) systematic
(C) representative
(D) non-representative

Q4. Which pair correctly matches symbols to their homes?

(A) $\bar{x}$ and s — population; μ and σ — sample
(B) both pairs describe the sample
(C) μ and σ — population; $\bar{x}$ and s — sample
(D) both pairs describe the population

Q5. A population has $\sigma = 15$. For samples of size $n = 25$, the standard error of the mean is:

(A) 15
(B) 0.6
(C) 3
(D) 5

Q6. The Central Limit Theorem says that as n grows, the sampling distribution of $|x|$:

(A) matches the population's shape exactly
(B) becomes normal regardless of the population's shape
(C) widens without limit
(D) becomes uniform

Q7. The null hypothesis in "the machine's mean output is 100 units" being tested is:

(A) $\mu \neq 100$
(B) $\mu > 100$
(C) $\mu = 100$
(D) $\mu < 100$

Q8. For a one-sample t-test with a sample of 9 observations, the degrees of freedom are:

(A) 9
(B) 10
(C) 7
(D) 8

Q9. A sample of 16 gives $|x| = 48$, $s = 4$, testing $H_0 : \mu = 50$. The t-statistic is:

(A) -2 **(B)** -0.5
(C) -8 **(D)** 2.131

Q10. In the same test, the two-tailed 5% critical value for 15 df is 2.131. The verdict is:

(A) reject H_0 — significant **(B)** fail to reject H_0 — not significant
(C) accept H_1 as proven true **(D)** increase the sample first

Detailed Solutions

Sol. Q1. Computed from the ENTIRE population (all 2,000 students), the number is a fixed population constant — a parameter. Statistics belong to samples; the standard error lives on sampling distributions.

Answer: (B)

Sol. Q2. The jump is $k = \frac{N}{n} = \frac{200}{20} = 10$: random start in 1–10, then every 10th unit — 10 well-spread picks from one random act.

Answer: (D)

Sol. Q3. Convenience picking ignores everyone outside that room — a structurally skewed, non-representative sample. The flaw is in the choosing, not the arithmetic.

Answer: (D)

Sol. Q4. Greek for the population (μ, σ — parameters), Latin for the sample ($\bar{x}$, s — statistics). Option (A) is the entire chapter's bookkeeping worn inside out.

Answer: (C)

Sol. Q5. SE $= \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{25}} = \frac{15}{5} = 3$. Option (A) forgot the root entirely; option (B) divided by n^2-thinking. The $\sqrt{n}$ scale is the whole computation.

Answer: (C)

Sol. Q6. That is the theorem's punchline: whatever the population looks like, the means of large samples pile into a normal shape centred at μ. The population's own shape fades as n grows.

Answer: (A)

Sol. Q7. The null hypothesis carries the status quo WITH equality: $\mu = 100$. The strict $\neq$, $>$, $<$ forms belong to the alternate hypothesis.

Answer: (C)

Sol. Q8. Degrees of freedom: $n - 1 = 8$ — one degree is spent estimating the sample mean before the spread can be judged.

Answer: (D)

Sol. Q9. Standard error: $\frac{s}{\sqrt{n}} = \frac{4}{4} = 1$. Then $t = \frac{48-50}{1} = -2$. The sign records the direction (lighter); the magnitude faces the table.

Answer: (A)

Sol. Q10. $|-2| = 2 < 2.131$: the result sits inside the acceptance region — fail to reject H_0. "Accept H_1 as proven" is double-banned: wrong direction and proof language.

Answer: (B)

Wrong-Answer Clinic

If you picked this	The thinking slip	The lasting fix
(A) in Q1	"A computed average = statistic" reflex	Ask WHERE the number came from: whole population = parameter
(A) in Q2	Reused n as the interval	k = N/n — population over sample, always
(C) in Q3	Convenient read as "good enough"	Representative = mirrors spread; convenience = bias
(A) in Q5	Dropped the square root	Write $\frac{\sigma}{\sqrt{n}}$ fully before substituting
(A) in Q8	Counted observations as degrees	Subtract 1 — the mean costs a degree
(A) in Q10	Read $2 < 2.131$ as "big enough"	Beyond the critical value rejects; inside keeps — the boundary is the boundary

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